Research methodology - Scientific papers

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Sep. 19, 2022, ELTE

Contents

1. Scientific texts
2. Scientific paper

Scientific texts

Based on: Antoni Martínez Ballesté: Writing Scientific Papers.

From manuscripts (XVII century) to online (open access) publications.

Scientific texts:

- Research paper;
- Report;
- Survey paper;
- Position paper;
- Letter;
- Scientific book;
- Book of proceedings, book of abstracts;
- Ph.D. thesis; master thesis;

Scientific paper

The **IMR**a**D** structure of a scientific research paper:

- Authors with affiliation, ORCID, email address;
- Title (identifies the new result);
- Abstract (briefly summarise - IMRaD itself);
- Keywords, ACM classification;
- **I**ntroduction (topic, research question, goal, background, related work);
- **M**ethods (new proposal);
- **R**esults and validation (section assessing the method and/or comparing the new proposal with existing ones);
- **D**iscussion (summary of achivements, further work);
- Acknowledgements;
- Bibliography;
- Appendices;

New Methods and Results

Research paper is based on a new method and result or improvement of an exsisting one (beyond the state of the art).

- An algorithm;
- A communication protocol;
- An architecture - including algorithms and communication protocols;
- Theoretical results (theorems, lemmas and corollaries);

Title

Complete, original, unique, relevant on new result.

Examples (from Scientific Writing Modul):

- *Privacy Preserving Techniques* - broad topic;
- *Privacy Preserving Techniques in Statistical Databases* - more focused, but survey or new result?
- *A New Method for Privacy Preservation in Statistical Databases* - research paper, result undefined;
- *A New Method for Privacy Preservation in Statistical Databases Based on Improving Microaggregation* - includes reference to own proposal
- *An MDAV Based Approach for Near-Optimal Microaggregation in Numerical Databases*
- *Location Privacy Through Users' Collaboration: A Distributed Pseudonymizer* - compound, clear: identifies new result and method;

Authors - roles

- New idea proposer;
- Doing experiments, implementing prototype;
- Literature research;
- Writing the text;
- Internal reviewer - in acknowledgements;

Order of authors, corresponding author;

Acknowledgements - supported by;

Abstract

50-300 words summary of the topic and the main achievements - freely accessible.

- In present tense;
- summarises the aim of the work;
- describes the methodology;
- summarises the results;

May be followed by keywords and classification in ACM Computing Classification System: https://dl.acm.org/ccs , https://cran.r-project.org/web/classifications/ACM.html

e.g.: D.1.1: Applicative (Functional) Programming
D.2: SOFTWARE ENGINEERING
Keywords: static analysis, refactoring

Introduction

- Introduce the topic (depending of the audience);
- Problem to be solved and main goals;
- Background concepts;
- Previous Work/Related Work/State of the Art
- Classification of the previous work;
- References: bibliographic citation with main features and shortcomings;

Cited by - towards survey of the state of the art

Google Scholar

Codecompass: an open software comprehension framework for industrial usage
Z Porkoláb, T Brunner, D Krupp, M Csordás - Proceedings of the 26th Conference on Program ..., 2018
Idézetek száma: 22 Kapcsolódó cikkek Mind a(z) 4 változat

Qilin: A New Framework For Supporting Fine-Grained Context-Sensitivity in Java Pointer Analysis
D He, J Lu, J Xue - 36th European Conference on Object ..., 2022 - drops.dagstuhl.de
Existing whole-program context-sensitive pointer analysis frameworks for Java, which were open-sourced over one decade ago, were designed and implemented to support only ...
☆ Mentés 99 Hivatkozás Idézetek száma: 1 Mind a(z) 6 változat »

99 A large-scale study of usability criteria addressed by static analysis tools ISSTA 2022
Marcus Nachtigall, Michael Schlichtig, E. Bodden

Presentation of new results

- algorithm: textual, pseudocode, flow-chart, structogram;
- protocol: content and flow of messages, sequence diagrams;
- architecture: figure with components and interactions;
- theoretical results: definitions, lemmas, theorems, proofs, formulae in a structured order;

Experiments, simulations, tests

- presentation of data: tables and charts - with captions;
- focus on most significant results, do not include redundant or unneccessary data;
- units / statistical notations;
- choose appropriate type of chart;
- axes, title, legend;
- monochrome readability;

Discussion and conclusion

- goals achieved?;
- summary in past tense - based on the text of the paper;
- comparison of results with the related work;
- new research questions, further work;

Bibliography, References

- references of scientific texts, web pages cited in the text;
- different citation and bibliography styles, e.g. [1], [1,3], [1,5-9], [Dij76], (Dijsktra 76), (Kiss et al., 94b).
- use standard biblioghraphy styles `documentclass{llncs}`;
- use BibTeX based on bibtex records collected in bib files;
- cited items are collected
- make file, make all, multiple compilation

```
all:
pdflatex main.tex
bibtex main.aux
pdflatex main.tex
pdflatex main.tex
rm -f *.aux *.log *.bbl *.blg *.out
clean: rm -f *.aux *.log *.bbl *.blg *.pdf *.out
```

Acknowledgements and Appendices

- for funding - may be differ for the authors;
- \thanks parancs - title, author - footnote;
- Example: "This research was supported by Thematic Excellence Pro- gramme, Industry and Digitization Subprogramme, NRDI Office, 2019.";
- Anonymous reviewers, and personnel who have collaborated;
- "The author whishes to thank" for programming the simulations;
- Appendices contain complementary information for the body of the paper;
- "See Appendix 3 for the ...";

Style of writing

- well-structured text (IMRaD, sections, subsections);
- Scientific writing in English;
- Do not use synonyms;
- No ambiguity;
- Personal or impersonal style (passive) - same style - parallelism;
- Short sentences;
- Abbreviations and acronyms;
- Abbreviations should be introduced;
- "he/she", they
- Latin abbreviations: e.g. (for example), i.e. (in other words),
- Numbers - numbers below 10 are spelled out;
- Numbers beginning a sentence are spelled out;

Structure of presentations

- use LaTeXBeamer;
- 8-15 lines/slide - do not use small letters;
- Structure of the presentation: IMRaD, title page, outline, sections, titles, enumerations and itemize;
- At most one silde per minute - usually 15 minutes + 5 minutes Q&A;
- 20-40 words/slide;
- Items: phrases, few words, short sentences;
- Columns, multicolumns - use graphics - left of the text;
- Figures - same size of letters, vector graphics, few colors;
- Citations - name of the author: e.g. [Dijkstra 76];
- Prepare additional sildes for questions (details);
- Plain text vs formulas;
- Head and foots - short version of titles within [ ];
- Slide numbering, e.g. 12/15 or bullets in head;

Presentations at workshops, conferences

- Sessions, session chairs;
- Time limits;
- Training, trial talk;
- Repeat the question, ask if answered well;
- All questions and remarks are useful;